MAIE5102 Homework Assignment 1

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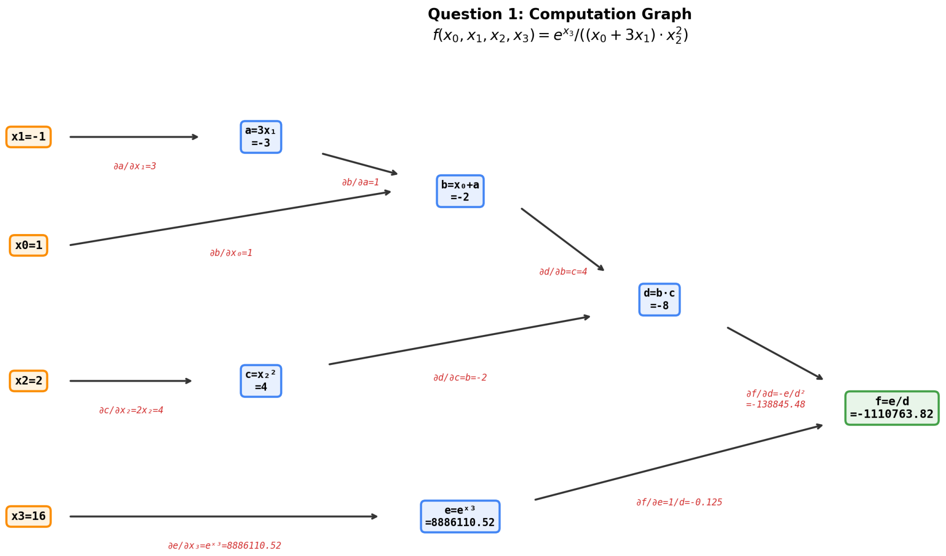
Question 1: Computation Graph (20 points)

Function: $f(x_0, x_1, x_2, x_3) = e^{x_3} / ((x_0 + 3x_1) \cdot x_2^{\,2})$

Given: $x_0=1$, $x_1=-1$, $x_2=2$, $x_3=16$

Computation Graph:



Forward Pass — Intermediate Outputs:

Node	Expression	Value
a	$3 \times x_1 = 3 \times (-1)$	–3
b	$x_0 + a = 1 + (-3)$	–2
c	$x_2^{\,2} = 2^2$	4
d	$b \times c = (-2) \times 4$	–8
e	$\exp(x_3) = \exp(16)$	8,886,110.52
f	e / d = 8,886,110.52 / (–8)	–1,110,763.82

Backward Pass — Gradients (Chain Rule):

Gradient	Derivation	Value
∂f/∂f	1	1

$\partial f/\partial e$	$1/d = 1/(-8)$	–0.125
$\partial f/\partial d$	$-e/d^2 = -8{,}886{,}110.52 / 64$	–138,845.48
$\partial f/\partial x_3$	$(\partial f/\partial e) \cdot \exp(x_3) = \exp(x_3)/d$	–1,110,763.82
$\partial f/\partial b$	$(\partial f/\partial d) \cdot c = (-138{,}845.48) \times 4$	–555,381.91
$\partial f/\partial c$	$(\partial f/\partial d) \cdot b = (-138{,}845.48) \times (-2)$	277,690.95
$\partial f/\partial x_0$	$(\partial f/\partial b) \cdot (\partial b/\partial x_0) = (-555{,}381.91) \times 1$	–555,381.91
$\partial f/\partial x_1$	$(\partial f/\partial b) \cdot (\partial b/\partial a) \cdot (\partial a/\partial x_1) = (-555{,}381.91) \times 3$	–1,666,145.72
$\partial f/\partial x_2$	$(\partial f/\partial c) \cdot (\partial c/\partial x_2) = 277{,}690.95 \times 2x_2 = 277{,}690.95 \times 4$	1,110,763.82

Final Gradients Summary:

• $\partial f/\partial x_0 = -555{,}381.91$

• $\partial f/\partial x_1 = -1{,}666{,}145.72$

• $\partial f/\partial x_2 = 1{,}110{,}763.82$

• $\partial f/\partial x_3 = -1{,}110{,}763.82$

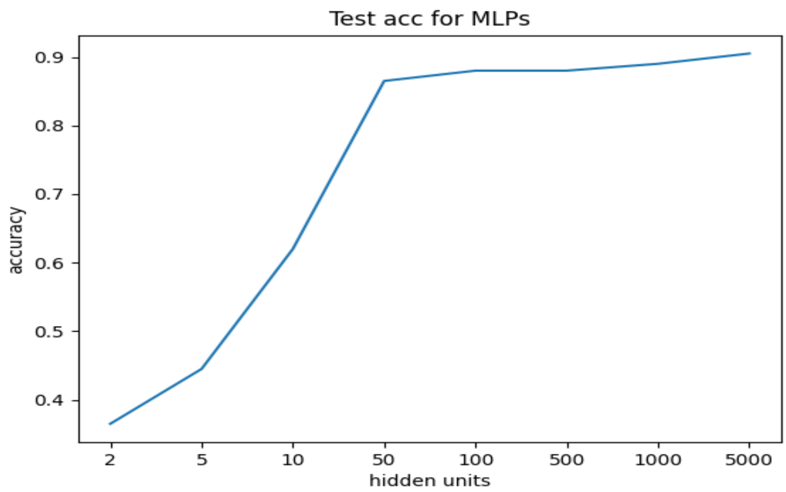
(Verified with PyTorch autograd — results match.)

Question 2: 2-Cluster — Observations (40 points)

After training single-hidden-layer MLPs with hidden units of 2, 5, 10, 50, 100, 500, 1000, and 5000, the following observations are made:

Test Accuracy vs. Number of Hidden Units:

Hidden Units	2	5	10	50	100	500	1000	5000
Accuracy	36.5%	44.5%	62.0%	86.5%	88.0%	88.0%	89.0%	90.5%



Observations:

1. Accuracy increases with hidden units: As the number of hidden units increases from 2 to 5000, test accuracy improves significantly — from 36.5% to 90.5%. This demonstrates that higher model capacity enables the MLP to learn more complex decision boundaries.

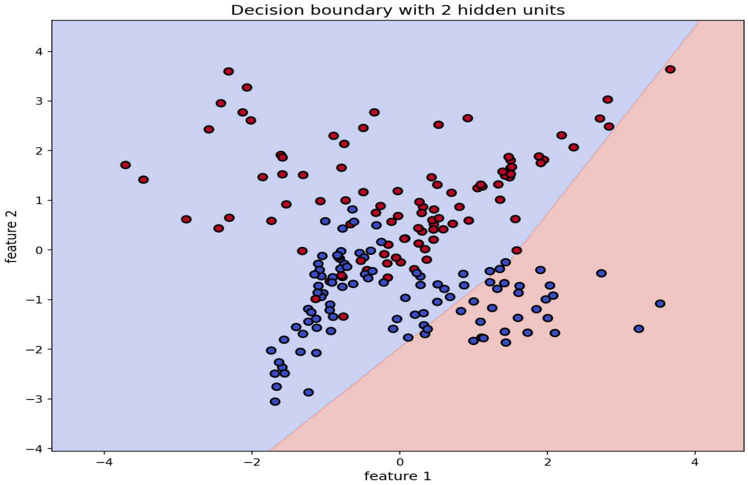
2. Rapid improvement in the low range: The most dramatic accuracy gains occur between 2 and 50 hidden units (36.5% → 86.5%), a nearly 50 percentage point improvement. This indicates that the 2-cluster classification task requires a moderate level of model capacity to separate the two classes.

3. Diminishing returns at higher capacity: Beyond 100 hidden units, accuracy improvements become marginal (88.0% → 90.5%). The accuracy curve plateaus, suggesting that additional hidden units provide diminishing returns once the model has sufficient capacity to capture the decision boundary.

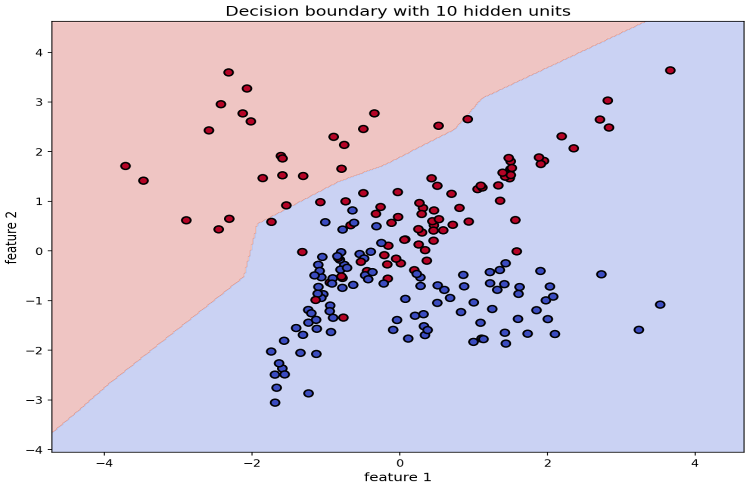
4. Decision boundary complexity: With very few hidden units (2 or 5), the decision boundaries are nearly linear and cannot separate the two clusters effectively. As hidden units increase (50+), the boundaries become increasingly non-linear and curved, closely conforming to the shape of the data clusters. With 1000 or 5000 units, the boundaries are highly detailed and smooth.

5. No clear overfitting observed: Even with 5000 hidden units (far exceeding the data complexity), the test accuracy continues to improve slightly rather than decrease, indicating that the model generalizes well within 30 training epochs. However, with more epochs, overfitting could potentially occur for very large models.

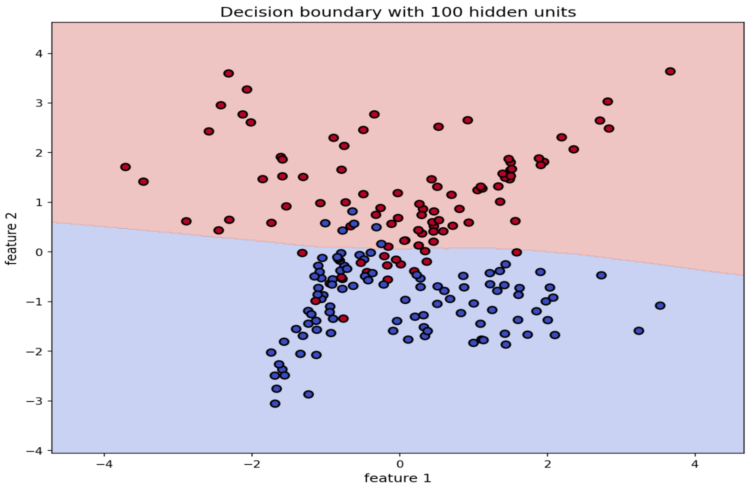
Decision Boundary Visualizations:



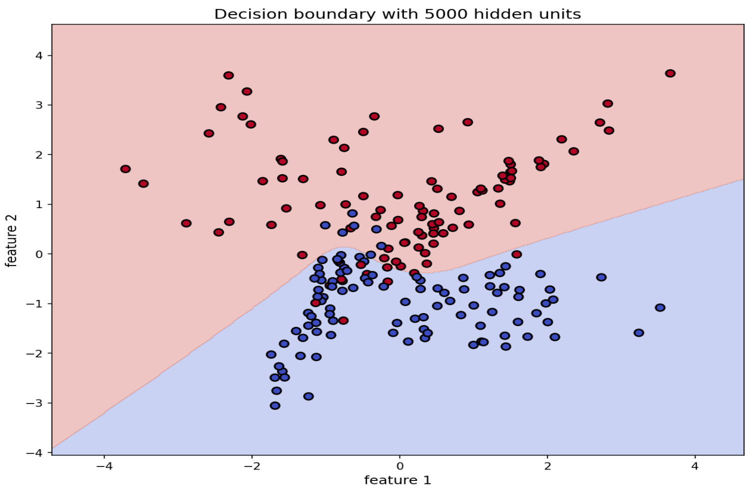
Decision boundary with 2 hidden units



Decision boundary with 10 hidden units



Decision boundary with 100 hidden units



Decision boundary with 5000 hidden units